We create chemistry

Safety data sheet

1. Identification

Product identifier

Sodium Nitrite FG 0,6% SiO2

Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: food additive(s)
Recommended use: Raw material, Intermediate, food additive(s), corrosion inhibitor, Surface treatment agent

Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY
Division Monomers

Telephone: +49 621 60 42737
E-mail address: pss.monomers@basf.com

Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

2. Hazards Identification

Classification of the substance or mixture

According to UN GHS criteria

Ox. Sol. 2
Acute Tox. 3 (oral)
Eye Dam./Irrit. 2A
Aquatic Acute 1

M-factor acute: 1

For the classifications not written out in full in this section the full text can be found in section 16.

Label elements

Globally Harmonized System (GHS)

Pictogram:

Signal Word:
Danger

Hazard Statement:

H319	Causes serious eye irritation.
H301	Toxic if swallowed.
H400	Very toxic to aquatic life.
H272	May intensify fire; oxidizer.

Precautionary Statements (Prevention):

P273	Avoid release to the environment.
P280	Wear protective gloves and eye protection or face protection.
P283	Wear fire resistant or flame retardant clothing.
P280	Wear eye and face protection.
P210	Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P270	Do not eat, drink or smoke when using this product.
P264	Wash contaminated body parts thoroughly after handling.
P221	Take any precaution to avoid mixing with combustibles ...
P220	Keep away from clothing and other combustible materials.

Precautionary Statements (Response):

P310	Immediately call a POISON CENTER or physician.
P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P301	IF SWALLOWED:
P330	Rinse mouth.
P391	Collect spillage.
P337	If eye irritation persists:
P311	Call a POISON CENTER or physician.
P370 + P378	In case of fire: Use water spray for extinction.

Precautionary Statements (Storage):

P405	Store locked up.
P420	Store separately.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.

According to UN GHS criteria

Hazard determining component(s) for labelling: Sodium nitrite

Other hazards

According to UN GHS criteria

No specific dangers known, if the regulations/notes for storage and handling are considered.

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

3. Composition/Information on Ingredients

Substances

Not applicable

Mixtures

Chemical nature

Sodium nitrite
NaNO2

Contains: anticaking agent, free flowing agent

Hazardous ingredients (GHS)
According to UN GHS criteria

Sodium nitrite

Content (W/W): >= 75 % - <= 100 %	Ox. Sol. 2
CAS Number: 7632-00-0	Acute Tox. 3 (oral)
EC-Number: 231-555-9	Eye Irrit. 2A
	Aquatic Acute 1
	M-factor acute: 1
	H272, H319, H301, H400

For the classifications not written out in full in this section the full text can be found in section 16.

4. First-Aid Measures

Description of first aid measures

First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Immediately wash thoroughly with soap and water, seek medical attention.

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far

Hazards: Risk of pulmonary edema. Symptoms can appear later. Danger of methaemoglobin formation after ingestion.

Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Extinguishing media

Suitable extinguishing media:
water spray

Unsuitable extinguishing media for safety reasons:
ABC powder, carbon dioxide

Special hazards arising from the substance or mixture

nitrogen oxides
The substances/groups of substances mentioned can be released in case of fire. Has a fire-promoting effect due to release of oxygen.

Advice for fire-fighters

Special protective equipment:
Wear a self-contained breathing apparatus.

Further information:
Substance/product is an oxidizing agent and can supply oxygen to stimulate or accelerate the combustion of organic or other combustible substances/products.

6. Accidental Release Measures

Personal precautions, protective equipment and emergency procedures

Use breathing apparatus if exposed to vapours/dust/aerosol. Avoid contact with eyes.

Environmental precautions

Do not discharge into the subsoil/soil. Do not discharge into waterways or sewer systems without proper authorization.

Methods and material for containment and cleaning up

For residues: Pick up with suitable appliance and dispose of.

7. Handling and Storage

Precautions for safe handling

Keep container tightly sealed. Breathing must be protected when large quantities are decanted without local exhaust ventilation. Processing machines must be fitted with local exhaust ventilation. Protect against moisture. Protect against heat. Do not mix with combustible substances. Handle in accordance with good industrial hygiene and safety practice.

Protection against fire and explosion:
The substance/product is non-combustible. Has a fire-promoting effect due to release of oxygen. Where required Prevent electrostatic charge - sources of ignition should be kept well clear - fire extinguishers should be kept handy.

Conditions for safe storage, including any incompatibilities

Segregate from oxidizable substances. Segregate from acids. Segregate from ammonium salts.

Suitable materials for containers: Carbon steel (Iron), Stainless steel 1.4541, Stainless steel 1.4571, High density polyethylene (HDPE), Low density polyethylene (LDPE), rubberized

Further information on storage conditions: Keep container tightly closed and in a well-ventilated place. This product is classified as a dangerous substance for storage. The authority permits and storage regulations must be observed. Keep away from food, drink and animal feeding stuffs.

Specific end use(s)

See exposure scenario(s) in the attachment to this safety data sheet.

8. Exposure Controls/Personal Protection

Control parameters

Components with occupational exposure limits

7632-00-0: Sodium nitrite

Exposure controls

Personal protective equipment

Respiratory protection:
Breathing protection if dusts are formed. Particle filter with high efficiency for solid and liquid particles (e.g. EN 143 or 149, Type P3 or FFP3).

Hand protection:
Chemical resistant protective gloves (EN ISO 374-1)
Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):
polyvinylchloride (PVC) - 0.7 mm coating thickness
nitrile rubber (NBR) - 0.4 mm coating thickness
chloroprene rubber (CR) - 0.5 mm coating thickness
butyl rubber (butyl) - 0.7 mm coating thickness
fluoroelastomer (FKM) - 0.7 mm coating thickness
Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing. Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures
Handle in accordance with good industrial hygiene and safety practice. Do not breathe dust. Keep away from food, drink and animal feeding stuffs. No eating, drinking, smoking or tobacco use at the place of work. Take off immediately all contaminated clothing. Hands and/or face should be washed before breaks and at the end of the shift.

9. Physical and Chemical Properties

9.1. Information on basic physical and chemical properties

State of matter:	solid	
Form:	crystalline	
Colour:	white to slightly yellow	
Odour:	odourless	
Odour threshold:	Not determined due to potential health hazard by inhalation.	
Melting point:	280 °C	(other)
Boiling point:	(1.013,25 hPa) The substance / product decomposes therefore not determined.	
Flammability:	not highly flammable	(other)
Lower explosion limit:	For solids not relevant for classification and labelling.	
Upper explosion limit:	For solids not relevant for classification and labelling.	
Flash point:	Study scientifically not justified.	
Thermal decomposition:	> 320 °C Nitrogen monoxide, nitrogen dioxide, Disodium oxide	
pH value:	approx. 7 - 9 (100 g/l) The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.	
Viscosity, kinematic:	not applicable, the product is a solid	
Viscosity, dynamic:	Study scientifically not justified., not applicable, the product is a solid	
Solubility in water:	readily soluble, Formation of sediments	
Partitioning coefficient n-octanol/water (log Kow):	Study scientifically not justified.	
Vapour pressure:	Study scientifically not justified.	
Relative density:	2,17 (20 °C) Literature data.	
Density:	2,17 g/cm3 (20 °C) Information based on the main component/s.	(ISO 2811-3)
Relative vapour density (air):	The product is a non-volatile solid.	

Particle characteristics
Particle size distribution: 200 - 400 µm (D50, measured)

9.2. Other information

Information with regard to physical hazard classes

Explosives
Explosion hazard: Based on the chemical structure there is no indication of explosive properties.
Impact sensitivity: not shock-sensitive
Based on the chemical structure there is no shock-sensitivity.
Oxidizing properties
Fire promoting properties: Oxidizing., maximum burning rate equal or higher than reference mixture (UN Test O.1 (oxidizing solids))
Self-heating substances and mixtures
Self heating ability: It is not a substance capable of spontaneous heating.
Corrosion to metals

Corrosive effects to metal are not anticipated. - In the presence of water or moisture metal corrosion cannot be excluded.

Other safety characteristics

Radioactivity: not radioactive for transport purposes
Bulk density: 1.100 - 1.300 kg/m3 (other)
pKA:
Study scientifically not justified.
Hygroscopy: hygroscopic
Surface tension:
Based on chemical structure, surface activity is not to be expected.
Evaporation rate:
The product is a non-volatile solid.

10. Stability and Reactivity

Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals: Corrosive effects to metal are not anticipated. In the presence of water or moisture metal corrosion cannot be excluded.

Chemical stability

The product is chemically stable.

Possibility of hazardous reactions

Hazardous reactions in presence of mentioned substances to avoid.

The product is stable if stored and handled as prescribed/indicated. Reacts with organic substances.

Conditions to avoid

See SDS section 7 - Handling and storage.

Incompatible materials

Substances to avoid:
reducing agents, oxidizable substances, ammonium salts, amines, amine compounds, acids

Hazardous decomposition products

Hazardous decomposition products:
Disodium oxide
nitrogen oxides

11. Toxicological Information

Information on toxicological effects

Acute toxicity

Assessment of acute toxicity:
Of high toxicity after single ingestion. There is a risk of damage to the blood (methemoglobinemia) after a single uptake.

Experimental/calculated data:
LD50 rat (oral): 180 mg/kg

(by inhalation):Study scientifically not justified.

(dermal):Study scientifically not justified.

Irritation

Assessment of irritating effects:
Not irritating to the skin. Eye contact causes irritation.

Experimental/calculated data:
Skin corrosion/irritation rabbit: non-irritant (OECD Guideline 404)

Serious eye damage/irritation rabbit: Irritant. (OECD Guideline 405)

Respiratory/Skin sensitization

Assessment of sensitization:
There is no evidence of a skin-sensitizing potential.

Experimental/calculated data:
Study scientifically not justified.

Germ cell mutagenicity

Assessment of mutagenicity:
The data available on mutagenic action are not consistent.

Carcinogenicity

Assessment of carcinogenicity:
In long-term studies in rats and mice in which the substance was given by drinking-water, a carcinogenic effect was not observed. Under certain conditions nitrites can enhance the formation of nitrosamines in vivo. Nitrosamines are carcinogenic in animal studies.

Reproductive toxicity

Assessment of reproduction toxicity:
The results of animal studies gave no indication of a fertility impairing effect.

Developmental toxicity

Assessment of teratogenicity:
In animal studies the substance did not cause malformations. Animal studies gave no indication of a developmental toxic effect at doses that were not toxic to the parental animals. After the uptake of small doses toxicity to development will not be expected in humans.

Specific target organ toxicity (single exposure)

Assessment of STOT single:
There is a risk of damage to the blood (methemoglobinemia) after a single uptake.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
After repeated administration the prominent effect is damage of the blood (methemoglobin formation).

Aspiration hazard

No aspiration hazard expected.

12. Ecological Information

Toxicity

Assessment of aquatic toxicity:
Very toxic (acute effect) to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Toxicity to fish:
LC50 (96 h) 0,54 - 26,3 mg/l, Salmo gairdneri, syn. O. mykiss (other, Flow through.)

Aquatic invertebrates:
LC50 (96 h) 4,93 mg/l, aquatic crustacea (static)
Literature data.

EC50 (48 h) 15,4 mg/l, Daphnia magna (OECD Guideline 202, part 1, static)
The statement of the toxic effect relates to the analytically determined concentration.

Aquatic plants:
EC50 (72 h) > 100 mg/l (growth rate), Scenedesmus subspicatus (OECD Guideline 201, static)
The statement of the toxic effect relates to the analytically determined concentration.

Microorganisms/Effect on activated sludge:
EC10 (3 h) 210 mg/l, activated sludge, domestic (OECD Guideline 209, static)
The details of the toxic effect relate to the nominal concentration.

EC50 (48 h) 421 mg/l, protozoa (other, static)

Chronic toxicity to fish:
No observed effect concentration (31 d) 6,16 mg/l, Ictalurus punctatus, syn: I. robustus (other, Flow through.)

Chronic toxicity to aquatic invertebrates:
No observed effect concentration (80 d) 9,86 mg/l, aquatic crustacea (Daphnia test chronic, static)

Assessment of terrestrial toxicity:
No data available.

Persistence and degradability

Assessment biodegradation and elimination (H2O):
Not applicable for inorganic substances.

Elimination information:
not applicable

Assessment of stability in water:
According to structural properties, hydrolysis is not expected/probable.
Information on Stability in Water (Hydrolysis):
not applicable

Bioaccumulative potential

Assessment bioaccumulation potential:
Accumulation in organisms is not to be expected.

Bioaccumulation potential:
Because of the n-octanol/water distribution coefficient (log Pow) accumulation in organisms is not to be expected.

Mobility in soil

Assessment transport between environmental compartments:

Volatility: The substance will not evaporate into the atmosphere from the water surface.
Adsorption in soil: Adsorption to solid soil phase is not expected.

Other adverse effects

The substance is not listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

Additional information

Other ecotoxicological advice:
Do not allow to enter soil, waterways or waste water channels. Do not release untreated into natural waters. Inhibition of degradation activity in activated sludge is not to be anticipated during correct introduction of low concentrations.

13. Disposal Considerations

Waste treatment methods

Contact manufacturer regarding recycling.
Check for possible recycling.
Contact waste centre regarding recycling.

Contaminated packaging:
Contaminated packaging should be emptied as far as possible and disposed of in accordance with official regulations after being thoroughly cleaned.

14. Transport Information

Land transport

ADR

UN number or ID number:	UN1500
UN proper shipping name:	SODIUM NITRITE
Transport hazard class(es):	5.1, 6.1, EHSM
Packing group:	III
Environmental hazards:	yes
Special precautions for user:	Tunnel code: E

RID

UN number or ID number:	UN1500
UN proper shipping name:	SODIUM NITRITE
Transport hazard class(es):	5.1, 6.1, EHSM

Packing group:	III
Environmental hazards:	yes
Special precautions for user:	None known

Inland waterway transport

ADN

UN number or ID number:	UN1500
UN proper shipping name:	SODIUM NITRITE
Transport hazard class(es):	5.1, 6.1, EHSM
Packing group:	III
Environmental hazards:	yes
Special precautions for user:	None known

Transport in inland waterway vessel

Not evaluated

Sea transport

IMDG

UN number or ID number:	UN 1500
UN proper shipping name:	SODIUM NITRITE
Transport hazard class(es):	5.1, 6.1, EHSM
Packing group:	III
Environmental hazards:	yes Marine pollutant: YES
Special precautions for user:	EmS: F-A; S-Q

Air transport

IATA/ICAO

UN number or ID number:	UN 1500
UN proper shipping name:	SODIUM NITRITE
Transport hazard class(es):	5.1, 6.1
Packing group:	III
Environmental hazards:	No Mark as dangerous for the environment is needed
Special precautions for user:	None known

Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

15. Regulatory Information

Safety, health and environmental regulations/legislation specific for the substance or mixture

Not applicable

16. Other Information

Assessment of the hazard classes according to UN GHS criteria (most recent version)

M-factor acute: 1

Full text of classifications, hazard symbols and hazard statements, if mentioned in section 2 or 3:

Ox. Sol.	Oxidising solids
Acute Tox.	Acute toxicity
Eye Dam./Irrit.	Serious eye damage/eye irritation
Aquatic Acute	Hazardous to the aquatic environment - acute
Eye Irrit.	Eye irritation
H272	May intensify fire; oxidizer.
H319	Causes serious eye irritation.
H301	Toxic if swallowed.
H400	Very toxic to aquatic life.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.